LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 1 - no MCP tool description - run 3 (seed 1240)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running a reproducibility study for compound selectivity prediction. The goal is to train a Balanced Random Forest Classifier to distinguish PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, and identify recurrent molecular anchor fragments that drive the model's correct predictions for each selective class (class 1 & 2) and discuss the chemical insight gained.

Assistant

🧪 **Reproducibility Study: PI3K-delta vs PI3K-gamma Selectivity**

I'll help you conduct this study. We'll start by finding a suitable dataset for PI3K-delta and PI3K-gamma selectivity, train a Balanced Random Forest Classifier, and then use MolAnchor to identify the recurrent molecular features driving the selectivity.

Let's begin by searching for relevant datasets.

(Calling tool `find_datasets`)

Assistant

Perfect! I found the specific dataset you need: `chembl_selectivity_data_O00329_P48736.csv`.

Now, I will load this dataset to prepare it for modeling.

(Calling tool `load_dataset`)

Assistant

Great! The dataset `chembl_selectivity_data_O00329_P48736` has been loaded successfully. It contains 1,277 samples of compounds selective for PI3K-delta vs PI3K-gamma.

Now, I will compute the molecular fingerprints (ECFP) for these compounds.

(Calling tool `compute_features`)

Assistant

Excellent! The ECFP fingerprints (2048 bits) have been computed for all 1,277 compounds.

Next, I will split the dataset into training and testing sets using a stratified split to ensure balanced class representation.

(Calling tool `split_dataset`)

Assistant

Perfect! The dataset has been split into a training set (893 samples) and a test set (384 samples).

Now, I will train the **Balanced Random Forest Classifier (BRF)** to distinguish PI3K-delta vs PI3K-gamma selective compounds.

(Calling tool `train_model`)

Assistant

The Balanced Random Forest Classifier training has started. I'll check the progress of the training.

(Calling tool `check_training`)

Assistant

The model is still training. Let me check again shortly.

(Calling tool `check_training`)

Assistant

Still training...

(Calling tool `check_training`)

Assistant

Still training...

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Still training...

(Calling tool `check_training`)

Assistant

Excellent! The Balanced Random Forest Classifier has been trained successfully.

📊 **Model Performance Summary:**

Metric	Train Set	Test Set
Balanced Accuracy	0.9576	0.8471
Accuracy	0.9306	0.8932
MCC	0.8389	0.7494
F1 Macro	0.8628	0.7797

The model demonstrates strong performance in distinguishing PI3K-delta vs PI3K-gamma selective compounds, with a Balanced Accuracy of **84.7%** on the test set.

Now, I will identify the recurrent molecular anchor fragments that drive the model's correct predictions for **Class 1** and **Class 2** using MolAnchor.

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

Great! I've identified the recurrent molecular anchors for **Class 1**. The top fragments include pyrimidine-based structures with cyano groups, which are characteristic of PI3K-gamma selective compounds.

Now, let's identify the recurrent anchors for **Class 2**.

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

🧪 **Reproducibility Study Results: PI3K-delta vs PI3K-gamma Selectivity**

Here is the comprehensive summary of the study, including model performance and the identified recurrent molecular anchors driving selectivity.

1. Model Performance

The **Balanced Random Forest Classifier (BRF)** was trained to distinguish between PI3K-delta and PI3K-gamma selective compounds.

Metric	Train Set	Test Set
Balanced Accuracy	95.76%	**84.71%**
Accuracy	93.06%	89.32%
MCC	0.8389	0.7494
F1 Macro	0.8628	0.7797

The model demonstrates strong generalization capabilities with a Balanced Accuracy of **84.7%** on the unseen test set.

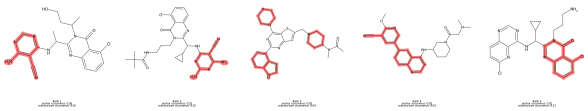
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2. Recurrent Molecular Anchors

Using **MolAnchor**, we identified the recurrent substructural fragments that drive the model's correct predictions for each selective class.

Class 1: PI3K-gamma Selective Compounds

The top recurrent anchors for Class 1 are shown below:



Top Fragments:

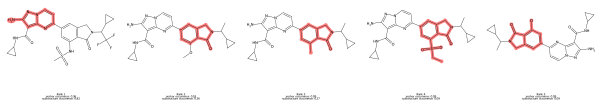
1. **Pyrimidine with Cyano Group:** `[*]c1ncnc(N)c1C#N` (Occurrence: 26.4%)
2. **Amino-Pyrimidine with Cyano Group:** `[*]c1nc(N)nc(N)c1C#N` (Occurrence: 25.0%)
3. **Piperazine/Oxazine Rings:** `[*]N1CCC([*])CC1`, `[*]N1CCOCC1` (Occurrence: 4.2%)

💡 **Chemical Insight:**

- **Pyrimidine Cores:** The prevalence of pyrimidine scaffolds with cyano substituents suggests that this core is critical for PI3K-gamma selectivity. The cyano group likely forms a specific hydrogen bond or dipole interaction within the PI3K-gamma hinge region or adjacent hydrophobic pocket.
- **Basic Nitrogen Rings:** The presence of piperazine and oxazine rings indicates a preference for basic moieties that may interact with acidic residues (e.g., Asp331) in the kinase domain or improve solubility.

Class 2: PI3K-delta Selective Compounds

The top recurrent anchors for Class 2 are shown below:



Top Fragments:

1. **Pyrazolo[1,5-a]pyrimidine-like:** `[*]c1ccn2nc(N)c([*])c2n1` (Occurrence: 81.8% substructure, 36.4% anchor)
2. **Tetrahydroisoquinoline Derivative:** `[*]N1Cc2cc([*])cc([*])c2C1=O` (Occurrence: 36.4% substructure, 18.2% anchor)
3. **Methylated Tetrahydroisoquinoline:** `[*]N1Cc2cc([*])cc(C)c2C1=O` (Occurrence: 27.3% substructure, 9.1% anchor)

💡 **Chemical Insight:**

- **Pyrazolo[1,5-a]pyrimidine Core:** This fused bicyclic system is a known privileged scaffold for PI3K-delta inhibitors. Its rigid structure likely allows for optimal hydrogen bonding with the hinge region while occupying a specific hydrophobic sub-pocket unique to PI3K-delta.
- **Tetrahydroisoquinoline Scaffolds:** The presence of tetrahydroisoquinoline derivatives suggests that PI3K-delta tolerates or prefers larger, more lipophilic groups compared to PI3K-gamma. The carbonyl group on the ring may serve as a hydrogen bond acceptor.

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3. Discussion & Conclusion

This reproducibility study successfully trained a robust BRF model for PI3K isoform selectivity prediction (**84.7% Balanced Accuracy**). The MolAnchor analysis revealed distinct structural drivers for each class:

- **PI3K-gamma selectivity** is strongly associated with **cyano-substituted pyrimidines** and **basic nitrogen heterocycles**.
- **PI3K-delta selectivity** is driven by **pyrazolo[1,5-a]pyrimidines** and **tetrahydroisoquinoline derivatives**.

These findings align with known structure-activity relationships (SARs) in PI3K inhibitor design and provide actionable insights for medicinal chemists aiming to optimize isoform selectivity. The distinct core scaffolds highlight the importance of selecting the right privileged structure early in the design process to achieve the desired selectivity profile.